Agentic RAG Chatbot

Multi-Format Document QA with Professional Interface

SLRIS

Key Features

- **Multi-agent architecture** with Model Context Protocol (MCP)
- **SLRIS-inspired branding** with authentic corporate design
- **Google Gemini LLM** integration with source context display
- **5 document formats:** PDF, DOCX, PPTX, CSV, TXT/MD
- **Vector search** with ChromaDB and semantic similarity

Technology Stack: Python · Flask · Streamlit · Google Gemini · ChromaDB · MCP

GitHub Repository: Agentic-RAG-Chatbot-for-Multi-Format-Document-QA-using-Model-Context-Protocol-MCP

https://github.com/45nivas/Agentic-RAG-Chatbot-for-Multi-Format-Document-QA-using-Model-Context-Protocol-MCP-

Agent-Based Architecture with MCP Integration

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┌──────────────────┐     ┌──────────────────┐     ┌──────────────────┐
│ CoordinatorAgent │◄──►│ IngestionAgent   │◄──►│ RetrievalAgent   │
│                  │     │                  │     │                  │
│ • Orchestrates   │     │ • Document       │     │ • Vector         │
│   workflow       │     │   parsing        │     │   embeddings     │
│ • Manages MCP    │     │ • Text chunking  │     │ • Similarity     │
│   messaging      │     │ • Multi-format   │     │   search         │
└──────────────────┘     └──────────────────┘     └──────────────────┘
            │                                                 │
            ▼                                                 ▼
┌──────────────────┐                                ┌──────────────────┐
│ LLMResponseAgent │                                │   ChromaDB       │
│ • Context-aware  │                                │ • Vector store   │
│   prompting      │                                │ • Embeddings     │
│ • Gemini LLM     │                                │ • Transformers   │
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MCP Features

- **Structured messaging** with trace IDs for traceability
- **Inter-agent communication** with standardized protocol
- **Error handling** and graceful failure recovery

System Flow Diagram (Message Passing)

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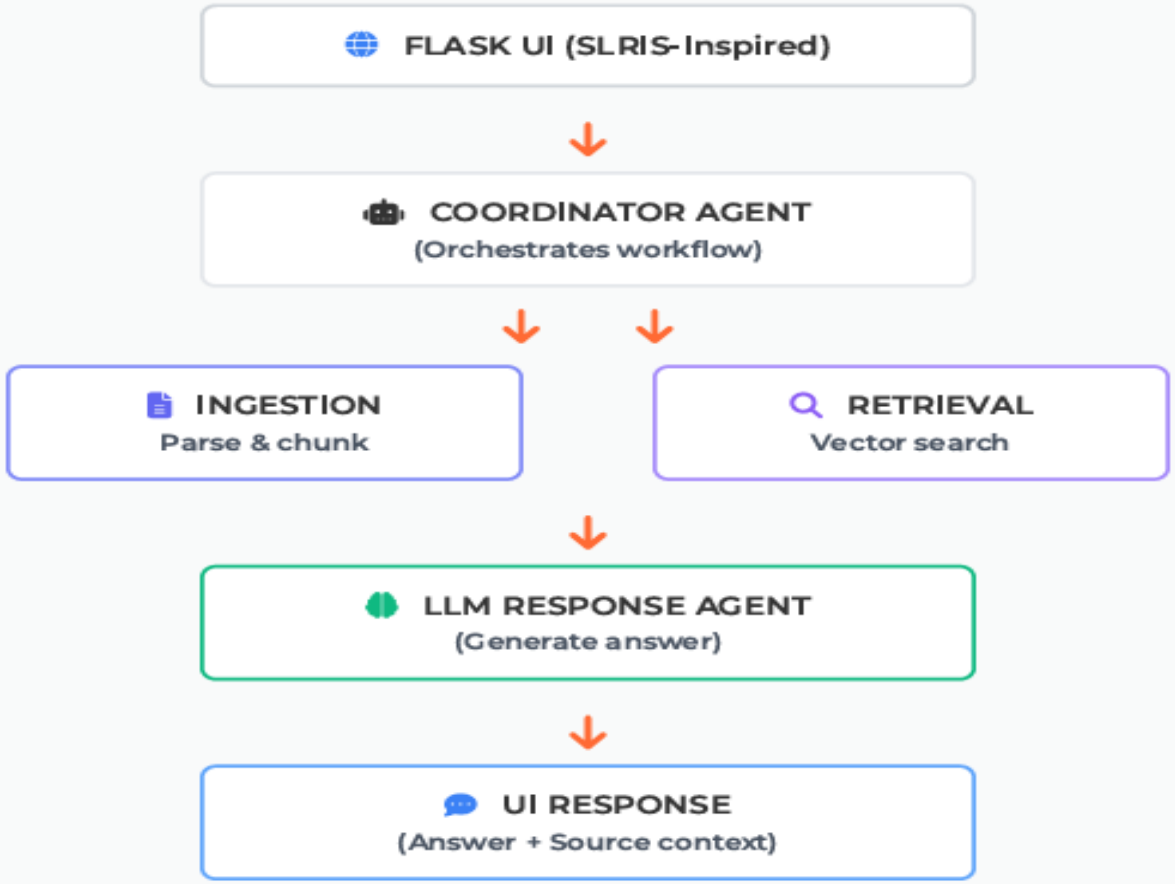
Upload: sales_review.pdf

Query: "What KPIs were tracked in Q1?"



Tech Stack & Components

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Frontend & UI

- **Flask:** Production web app
- **Streamlit:** Development interface
- **HTML/CSS/JS:** Responsive design
- **SLRIS Branding:** Authentic theme

Backend & AI

- **Google Gemini 1.5 Flash:** LLM
- **ChromaDB:** Vector database
- **Sentence Transformers:** Embeddings
- **Multi-format parsers:** Documents

Architecture

- **Model Context Protocol:** Custom MCP
- **4 Specialized Agents:** Modular design
- **Environment Variables:** Secure config
- **Session Management:** Persistence

Document Support

- **PDF:** PyPDF2 text extraction
- **DOCX:** Word document parsing
- **PPTX:** PowerPoint content
- **CSV/TXT/MD:** Structured & plain text

UI Screenshots & Features

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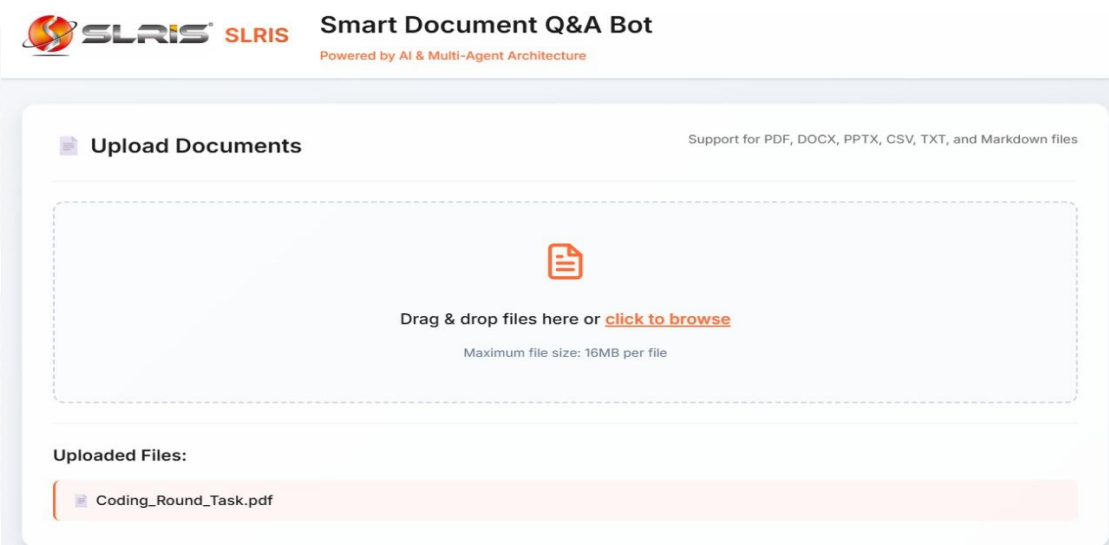
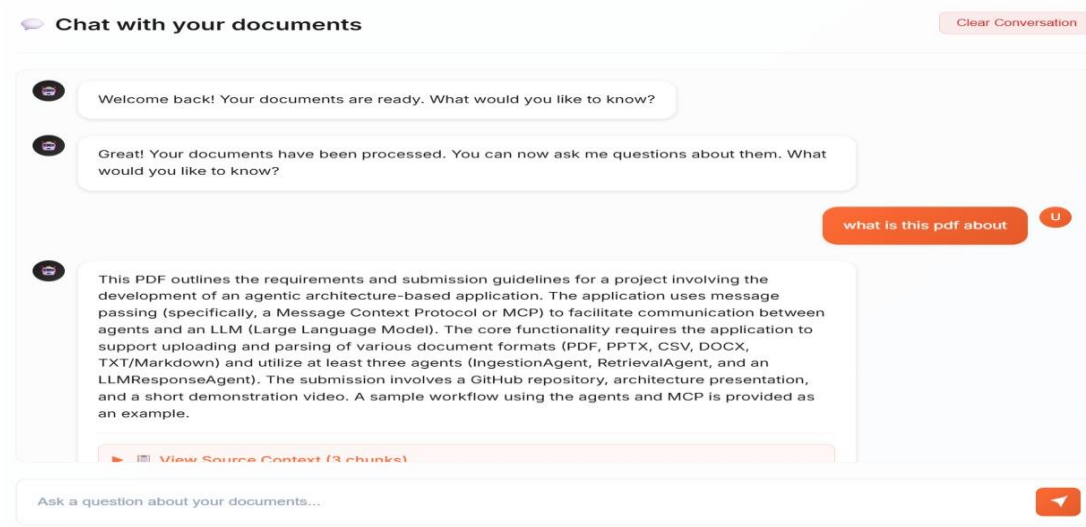
Professional Flask Interface



Source Context Display



Key Features Demonstrated

- ✓ Multi-format document upload
- ✓ Natural language questioning
- ✓ AI-powered responses with sources
- ✓ Professional SLRIS branding
- ✓ Responsive design
- ✓ Source context verification

Challenges & Future Scope

SLRIS

Challenges Faced

Google Gemini API Rate Limits
Free tier: 15 requests/minute causing 429 errors

Multi-Format Document Parsing
Consistent text extraction across PDF, DOCX, PPTX, CSV

MCP Protocol Design
Reliable inter-agent communication with traceability

SLRIS Brand Integration
Authentic corporate branding without trademark issues

Solutions Implemented

Intelligent Rate Limiting
User-friendly error messages with retry logic

Robust Document Processing
Specialized parsers with comprehensive error handling

Structured MCP Messaging
JSON-based protocol with trace IDs and validation

Professional UI/UX Design
"Inspired by" approach with legitimate design elements

Future Enhancements

- **Advanced Document Support:** OCR, Excel, Email processing
- **Production Scalability:** Microservices, load balancing
- **Enterprise Features:** SSO, RBAC, audit logging
- **Advanced Analytics:** Usage metrics, performance monitoring